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Polymerase trapping as the mechanism of H5 highly pathogenic avian influenza virus genesis

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Full article and list of author affiliations: https://doi.org/10.1126/science.adr6632

INTRODUCTION: Influenza viruses pose a continuous health threat to animals and humans worldwide. Outbreaks of highly pathogenic avian influenza viruses (HPAIVs) have a devastating impact on domestic and wild bird populations as well as on wild mammals. Moreover, HPAIVs present a health risk to humans through repeated spillover events, renewing concerns about the potential for a new pandemic. HPAIVs emerge from H5 and H7 low pathogenic variants upon the introduction of a furin-cleavable multibasic cleavage site (MBCS) in the hemagglutinin (HA) gene. The insertion of this site enables systemic virus spread in poultry, leading to severe and often lethal disease.

RATIONALE: Although MBCS acquisition was identified as the molecular hallmark of HPAIVs >40 years ago, the underlying molecular mechanisms have remained poorly understood. Recent advances in the structural and functional understanding of the influenza virus polymerase have allowed us to propose that MBCS-yielding insertions are caused by polymerase trapping by transient RNA structures formed by the entering and exiting template parts during replication of purine-rich sequences. To test this hypothesis, mutant H5 HAs with modified RNA structures or sequences were designed using an in-house bioinformatic tool predicting transient RNA template structures. A virus-free influenza replication system was used to detect all insertions, without constraints at the protein level or selection biases. The minimal replication unit of an H5 HPAIV, consisting of the polymerase, the nucleoprotein, and an H5 HA template, was reconstituted in mammalian and avian cells. A circular next-generation sequencing approach was developed to accurately determine rare insertions made by the influenza virus polymerase.

RESULTS: Rare duplication insertions, analogous to those seen in H5 HPAIV in nature, were observed in mutated purine-rich HA cleavage sites. Bioinformatic analysis predicted that these insertions could be the result of a putative transient RNA template structure forming during replication of complementary RNA genome (cRNA) to viral RNA genome (vRNA). The role of this structure was shown by either disrupting or stabilizing it, which had the anticipated effect on insertion frequencies. Its role was further confirmed with unidirectional replication systems, in which HPAIV-like duplication insertions mainly occurred when using cRNA and not vRNA templates. Comparable insertion frequencies and patterns were observed in mammalian and avian cells, suggesting a lack of host factor involvement. Introduction of H5-like cleavage site sequences and structures into an H6 HA resulted in MBCS-yielding insertions. Direct biophysical evidence was provided using cryo–electron microscopy, revealing polymerase stalling through base pairing of the incoming and outgoing template parts.

CONCLUSION: We provide empirical evidence of RNA structure involvement in HPAIV genesis. Our results show that MBCS acquisition is the result of the combination of insertion-prone RNA sequences and the formation of transient RNA structures. The interplay of these two factors might explain why HPAIV-yielding insertions are rare and restricted to two HA subtypes. Beyond influenza viruses, transient polymerase-trapping structures may also influence the replication or transcription of other negative-sense RNA viruses with similar polymerase entry and exit channel architecture. □

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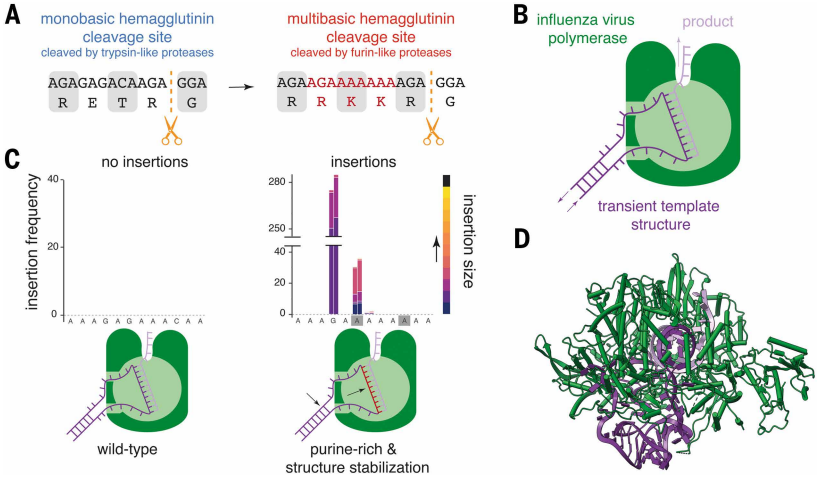

H5 highly pathogenic avian influenza virus genesis. (**A**) Highly pathogenic viruses emerge from low pathogenic variants upon the acquisition of a multibasic HA cleavage site. (**B**) Our model of influenza virus polymerase trapping by transient template structures. (**C**) Transient template structure stabilization and purine-rich HA cleavage site sequences promote multibasic cleavage site insertions. (**D**) Cryo–electron microscopy structure of polymerase trapping by template RNA structure.

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INFLUENZA

Polymerase trapping as the mechanism of H5 highly pathogenic avian influenza virus genesis

Mathis Funk[1]†, Monique I. Spronken[1], Roy M. Hutchinson[1], Benoit Arragain[2]‡, Pauline Juyoux[3], Theo M. Bestebroer[1], Anja C. M. de Bruin[1]§, Alexander P. Gultyaev[1,4], Ron A. M. Fouchier[1], Stephen Cusack[2], Aartjan J. W. te Velthuis[5], Mathilde Richard[1]*

Highly pathogenic avian influenza viruses (HPAIVs) derive from H5 and H7 low pathogenic avian influenza viruses (LPAIVs). Although insertion of a furin-cleavable multibasic cleavage site (MBCS) in the hemagglutinin gene was identified decades ago as the genetic basis for the LPAIV-to-HPAIV transition, the mechanisms underlying the occurrence of insertion are unknown. Here, we show that transient H5 RNA structures, predicted to trap the influenza virus polymerase on purine-rich sequences, drive nucleotide insertions, providing empirical evidence of RNA structure involvement in MBCS acquisition. Introduction of H5-like sequences and structures into an H6 hemagglutinin resulted in MBCS-yielding insertions. Our results show that nucleotide insertions that underlie H5 HPAIV emergence result from an RNA structure–driven diversity-generating mechanism, which could also occur in other RNA viruses.

Over the past 60 years, 54 independent highly pathogenic avian influenza virus (HPAIV) emergence events have been described (*1*, *2*). Outbreaks of HPAIV have devastating impacts on domestic and wild bird populations and present a global health risk from repeated spillover events into mammals, including humans (*3*). HPAIV emergence results from the acquisition of a multibasic cleavage site (MBCS) in the hemagglutinin (HA) gene, which encodes the influenza virus receptor binding and fusion protein that mediates virus entry into cells. Translation of the HA gene yields a precursor protein that is posttranslationally cleaved by host cell proteases, a step required for activation of the fusion potential of HA and therefore virus infectivity (*4*). Low pathogenic avian influenza virus (LPAIV) HAs have a monobasic cleavage site that can only be cleaved by tissue-restricted trypsin-like proteases. By contrast, the HPAIV MBCSs are cleaved by ubiquitous furin-like proteases, allowing systemic virus spread in terrestrial poultry (*1*). Although MBCS acquisition is the key genetic basis of LPAIV-to-HPAIV conversion and has been known since the 1980s (*5*, *6*), the underlying mechanisms have remained unknown. Genetic analysis of MBCSs from newly converted HPAIVs suggests three different origins. Two result from errors by the influenza virus RNA-dependent RNA polymerase (RdRp) during influenza virus RNA replication (*1*, *7*–*10*). These errors are either nucleotide substitutions leading to nonbasic to basic amino acid changes and/or insertions of nucleotides coding for basic amino acids, possibly by duplication of neighboring sequences. The third origin, observed in H7 only, results from nonhomologous recombination (NHR) with host or viral RNA (*11*–*15*), in which the involvement of the RdRp is unclear. In this study, we sought to elucidate the mechanism of MBCS acquisition by duplication, as it is the most likely mechanism underlying 38 of the 49 HPAIV conversion events that have been identified with known MBCS sequences and the majority of H5 LPAIV-to-HPAIV conversions (*1*, *2*).

The restriction of MBCS acquisition to H5 and H7 HAs is not due to incompatibilities at the protein level, as non-H5/H7 HAs can accommodate artificial MBCSs that can be cleaved by endogenous proteases (*16*–*20*). We and others have hypothesized that insertions at the HA cleavage site in H5 and H7 HAs might be caused by HA RNA sequence and/or structure (*7*, *8*, *10*, *21*–*27*). The functional influenza virus replication unit is a viral ribonucleoprotein (vRNP) complex comprising the RdRp [polymerase basic 2 (PB2), polymerase basic 1 (PB1), and polymerase acidic (PA) subunits], nucleoprotein (NP), and a template RNA, containing highly conserved 5′ and 3′ promoter sequences. Whereas vRNAs (negative-sense genomic influenza virus RNAs) and cRNAs (positive-sense copies of vRNA serving as replication intermediates) are bound by NP, the nonuniformity of NP coverage allows secondary RNA structures to form in cells and virions (*28*, *29*). Perdue *et al.* (*8*) have hypothesized that the RdRp can be stalled during replication by downstream RNA structures, leading to duplications. However, this model is incompatible with our previous observation that the insertions are located in the loop of conserved in silico predicted stem-loop structures (*21*). Recent insights from RdRp structures during transcription have revealed that the template entrance and exit channels are in close proximity (*30*, *31*), theoretically allowing the formation of double-stranded RNA structures between the entering and exiting template strands. These structures would prevent movement of the template into the active site, impeding replication (Fig. 1A), and would be transient because the entering and exiting RNA template parts brought into close proximity will shift as the RdRp moves from the 3′ to the 5′ end of the template RNA. By combining this understanding and the prediction of stem-loop RNA structures at the HA cleavage site, we have proposed a model in which the RdRp becomes trapped (*1*, *24*). In this model, a transient RNA structure is formed by the template during RNA replication, which encloses and traps the RdRp. This physical constraint on the RdRp causes it to stutter (replicate the same nucleotide repeatedly) or backtrack (sliding back along the template over several nucleotides before reinitiating replication), leading to an increase in duplication insertions. Previous studies have highlighted that viruses with adenine or uracil (A/U)–rich HA cleavage sites (cRNA and vRNA, respectively) are more prone to accumulate insertions and convert to HPAIV (*22*, *25*, *32*–*39*), indicating that sequence of the HA cleavage site also plays an important role. Therefore, we further hypothesized that the RdRp needs to be trapped by transient RNA structures on A/U-rich sequences to generate duplications and that both factors act synergistically. In another context, putative RdRp-trapping structures, coined t-loops, have recently been shown to interfere with the replication of short aberrant influenza virus mini viral RNAs (mvRNAs) (*40*), supporting the previously proposed model for insertions at the HA cleavage site (*1*, *24*).

The difficulty of reproducing the rare event of MBCS acquisition in laboratory settings has been a hindrance. In addition, the presence of highly repetitive sequences in MBCSs complicates the reliable detection of insertions, as these sequences are prone to insertions or deletions for many replication enzymes, such as reverse transcriptases and DNA polymerases needed in the process of RNA sequencing. In this study, we used an alternative approach in which a virus-free replication system was combined with customized circular sequencing (*41*) to accurately identify and quantify insertions introduced by the RdRp during RNA replication. A virus-free system allowed the characterization of the full RdRp insertional landscape free from protein translation

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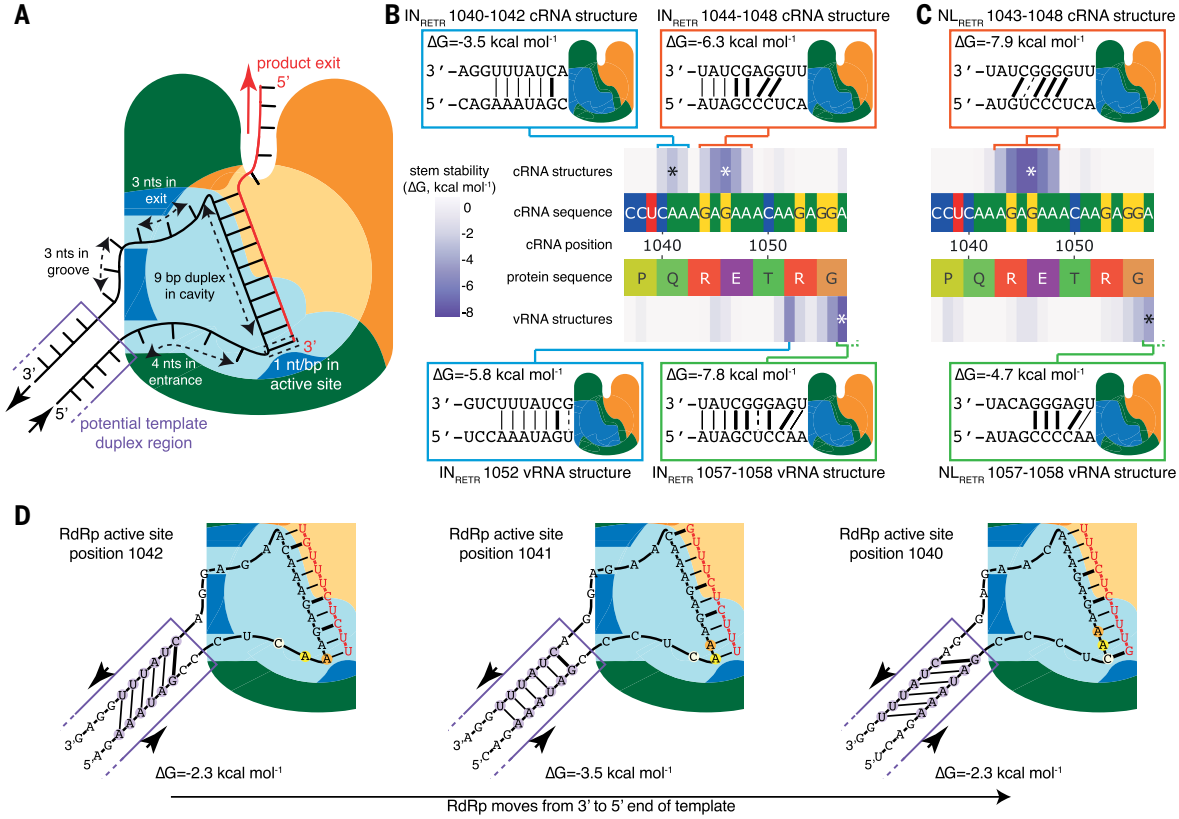

Fig. 1. The trapped RdRp model and predicted transient RNA structures at the HA cleavage site. (**A**) Schematic representation of the RdRp (PB2, orange; PB1, blue; PA, green), during replication of a template RNA (black) into a product RNA (red). The template threads through the RdRp [approximate lengths based on PDB ID 6SZU (*46*)] and exits adjacent to the template entrance, allowing potential base pairing. nts, nucleotides; bp, base pairs. (**B** and **C**) Predicted RdRp-trapping structures at the HA cleavage sites of A/Indonesia/5/2005 [IN_{RETR}, (B)] and A/mallard/Netherlands/3/1999 [NL_{RETR}, (C)]. The nucleotide sequence of the HA cleavage site (cRNA sense) and the corresponding amino acid sequence are colored by nucleotide or amino acid. Numbering starts at the first 5′ nucleotide of the segment. For each active site (+1) position, the free energy change ΔG (expressed as kilocalories per mole) of the RdRp-trapping structure is represented by the heatmaps according to the scale on the left. Lower ΔG values reflect higher RNA structure stability. Each transient RNA structure was named after the range of +1 positions for which identical base pairs were predicted to form. Structures only identified in IN_{RETR} are shown in blue frames, while structures predicted to form for analogous +1 positions in both IN_{RETR} and NL_{RETR} are shown in orange (cRNA) and green (vRNA) frames. Brackets on the heatmap indicate that the same structure is predicted to form for several +1 positions. The asterisk identifies the +1 position when the structure is at its most stable conformation, corresponding to the schematic representation. A thick line indicates a G-C, a thin line an A-U, and a dashed line a G-U base pair. (**D**) Detailed representation of the same cRNA structure predicted to form for +1 positions 1040–1042 on IN_{RETR} cRNA, using the same representation as (A).

and viral selection constraints. Our results show that two factors directly influence insertion at the H5 cleavage site: the formation of putative RdRp-trapping RNA structures and the template RNA sequence in the RdRp active site cavity, with sequences containing long A stretches being more insertion-prone.

Predicting RdRp-trapping RNA structures

As it is well established that insertions in H7 HAs often occur through NHR, in addition to potential duplication (*1*), we focused on investigating the molecular mechanism underlying insertions at H5 HA cleavage sites. We chose two model sequences: A/mallard/Netherlands/3/1999 HA (NL_X, where X is the amino acid sequence of the HA cleavage site) and A/Indonesia/5/2005 HA (IN_X). The former was chosen as its cleavage site nucleotide sequence, as well as the 30 nucleotides (nt) up- and downstream, corresponds to the consensus of viruses from the Africa-Eurasia-Oceania (AEO) H5 LPAIV lineage (*42*). The latter is a representative of H5 HPAIVs of the A/goose/Guangdong/1/1996 (Gs/Gd) lineage, which first emerged in 1996 and is still circulating in birds (*43*–*45*). HAs of Gs/Gd lineage viruses have acquired substitutions in regions flanking the HA cleavage site that stabilize predicted RNA structures (*21*). To mimic an LPAIV HA, the wild-type (WT) $IN_{RESRRKKR}$ MBCS was removed and replaced with the highly conserved consensus H5 LPAIV cleavage site nucleotide sequence, coding for the amino acid sequence RETR.

To investigate the impact of putative transient RNA structures on insertions, we designed an RdRp-trapping structure prediction tool. We modeled transient RNA structures instead of static structures as might exist in nonreplicating vRNPs (*21*, *24*, *25*) by using a sliding window approach, as described previously (*40*). The 20-nt RdRp footprint (*46*), corresponding to the RNA that is interacting with the RdRp, was slid along the RNA template sequence nucleotide by nucleotide, in both the cRNA and the vRNA sense. For each nucleotide in the RdRp active site, referred to as +1 positions hereafter, the free energy change (ΔG) of RNA structure formation between 10-nt-long RdRp-flanking regions on either side was determined while mimicking the close proximity of entering and exiting template strands (see methods and fig. S1). Predicted RdRp-trapping RNA structures at the HA cleavage site of IN_{RETR} and NL_{RETR} are shown in Fig. 1, B and C, and an interactive graph showing predictions for all HA sequences used in this manuscript is provided as data S1 (*47*). When predicted transient RNA structures contain unpaired nucleotides close to the RdRp, identical base pairs can form for different +1 positions (Fig. 1D). Differences in the number of unpaired nucleotides on each side of the RdRp might

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still lead to varying predicted stability, with symmetrical distributions being more stable.

Four main transient RNA structures were predicted to form in IN_{RETR}, whereas only two of them were predicted in NL_{RETR}. The structures found only in IN_{RETR} result from the aforementioned Gs/Gd lineage-specific substitutions (*21*), absent in H5 LPAIV HAs such as NL_{RETR}. The shared cRNA structure predicted to form for +1 positions 1043–1048 and 1044–1048 for NL_{RETR} and IN_{RETR}, respectively (referred to as the 1043–1048 structure hereafter), was more stable in NL_{RETR}, whereas the shared predicted vRNA structure for +1 positions 1057 and 1058 was more stable in IN_{RETR}. MBCS-yielding insertions have occurred in nature between positions 1041 and 1052 (*1*), encompassing the IN-specific structures and the shared cRNA structure.

Insertions are promoted by A/U stretches

To experimentally assess insertion frequencies observed at the HA cleavage site, we generated HA RNPs through transfections in two biological replicates, and vRNA-sense HA extracted from cells was sequenced using a custom circular sequencing method (*41*). We focused on insertions, as they are the most relevant for H5 MBCS acquisition. Insertion frequencies were corrected for non-RdRp background insertions and potential duplications were realigned to the most stable predicted RdRp-trapping structure, standardizing insertion placement and testing compatibility with our model (see methods). We first investigated the impact of HA cleavage site sequence on insertions. Insertions were rarely observed at the LPAIV IN_{RETR} and NL_{RETR} HA cleavage sites (Fig. 2A). We and others have previously suggested that H5 MBCS acquisition occurs after initial nucleotide substitutions in the genetically stable LPAIV RETR cleavage site (*22*, *36*, *42*). The RETR-encoding cleavage site sequence was replaced either by REKR, which has been previously suggested to be a stepping-stone for MBCS acquisition (*34*, *36*), or by RKKR, which has been shown to be insertion-prone (*22*, *25*, *32*, *48*) (all HA nucleotide sequences are provided in data S2). The REKR cleavage site sequence led to increased insertions, but at very low frequencies [IN_{REKR}, 0.8 to 1.1 per mil (‰); NL_{REKR}, 0.8 to 0.9‰; fig. S2A], whereas insertions were detected at much higher frequencies in RKKR-bearing HAs (Fig. 2B)—up to 30.5‰ for GA insertions at position 1044 (1044GA) in NL_{RKKR}. The insertions observed in the REKR- and RKKR-containing HAs were reminiscent of those seen in H5 MBCSs in nature, consisting of additional As and Gs between cRNA positions 1041 and 1052 (*1*) (data S3). Insertions that could be explained by a single duplication were sorted into three types: (i) single-nucleotide insertions, most commonly 1046A; (ii) homopolymer insertions, most commonly 1046AA or 1046AAA; and (iii) heteropolymer insertions, most commonly 1044GA or 1044GAA. The most common insertions for all three groups occurred in or adjacent to the 1045–1052 eight-A stretch. A minority of insertions could not be explained by direct duplications of adjacent sequences, and these were sorted into a fourth

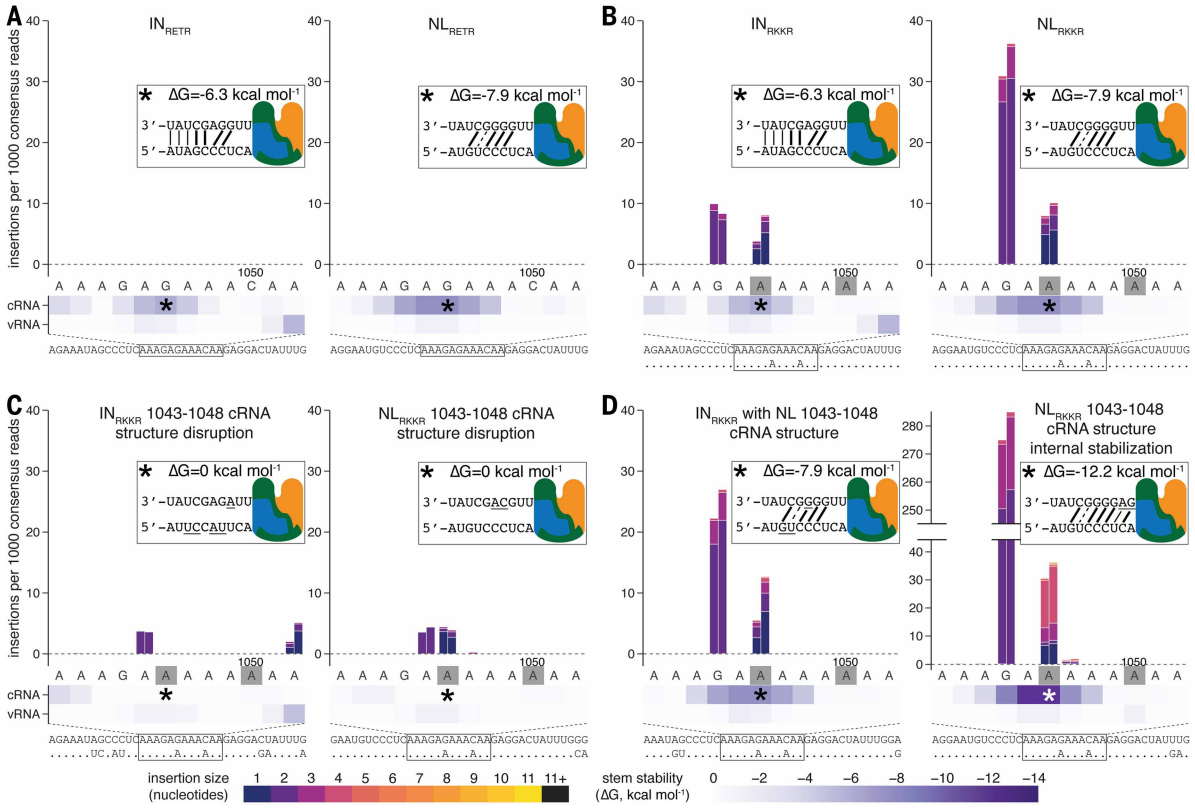

Fig. 2. The predicted 1043–1048 cRNA structure results in insertions in A/U-rich cleavage sites. Insertion frequencies observed in A/Indonesia/5/2005 (IN, left) and A/mallard/Netherlands/3/1999 (NL, right), with the consensus H5 LPAIV RETR cleavage site (**A**), with an RKKR cleavage site (**B**), with an RKKR cleavage site and substitutions disrupting the 1043–1048 cRNA structure (**C**), with RKKR cleavage site and substitutions stabilizing the 1043–1048 cRNA structure (**D**). Stacked bars show the number of insertions per 1000 consensus reads, background-corrected by subtracting frequencies observed in non-RdRp samples. The color of the bar reflects the size of the insertion (see the bottom legend). Results of two independent replicates are shown. The cRNA sense nucleotide sequence is indicated on the *x* axis. Heatmaps below the *x* axis indicate the stability (Δ*G*, expressed as kilocalories per mole) of predicted RdRp-trapping structures for each +1 position, according to the bottom heatmap scale. The sequence below the graphs shows all substitutions relative to the RETR-containing HA. Insets in each graph show a schematic representation of the RdRp-trapping structure predicted to form for the +1 position marked by an asterisk on the heatmap, using the same representation as in Fig. 1, with substitutions relative to the corresponding RETR-containing HA underlined. Uncorrected insertion frequencies for RdRp-containing and background samples are provided as an interactive figure in data S3 or available online (*47*).

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type: complex insertions (IN_{RKKR}, 0.6 to 0.6‰; NL_{RKKR}, 1.7 to 2.0‰). They may be explained by several consecutive insertion, deletion, and/or substitution events. Uncorrected insertion frequencies for each category for each template and replicate can be found in data S4. Some complex large insertions of low similarity with neighboring sequences most likely resulted from NHR. In total, 33 putative NHR events were observed across all samples (data S6). This included samples without RdRp, suggesting that NHR might occur independently of viral replication, for example, after RNA strand breakage in cells, as suggested for pestiviruses (*49*–*51*) and poliovirus (*52*). The majority of the inserted sequences showed identity with at least one human genome sequence via BLAST (Basic Local Alignment Search Tool), most commonly with ribosomal RNA (rRNA) sequences, which have previously been suggested to be the origin of NHR-derived insertions in H7 HA genes (*13*–*15*).

Heteropolymer insertions were more common than single-nucleotide insertions and homopolymer insertions in NL_{RKKR} (31.0 to 36.3‰, 4.6 to 5.3‰, and 2.2 to 3.3‰, respectively) and IN_{RKKR} (8.5 to 10.1‰, 2.4 to 5.1‰, and 1.0 to 2.5‰, respectively). Heteropolymer frequencies were found to vary little across 21 independent IN_{RKKR} biological replicates or when using different primers for RNA sequencing, whereas frequencies of other insertion types were more variable (fig. S3). Insertions of 2 nt were the most frequent, and frequencies declined steeply as sizes increased beyond 2 nt (fig. S3D). As a result, only 1.6 to 2.1‰ and 5.1 to 7.3‰ consensus reads presented in-frame insertions for IN_{RKKR} and NL_{RKKR}, respectively (data S5). The most common in-frame insertions for both HAs were 1044GAA heteropolymer insertions, resulting in an MBCS found in several newly converted HPAIVs in nature (*1*) (AGAAGAAAAAAAAGA; RRKKR). No correlation between consensus read numbers and insertion frequencies was observed (fig. S3E). Across all samples, background insertions were almost exclusively insertions of one A (data S3), consistent with RNA polymerase I (Pol I) errors (*53*). Background insertion frequencies increased when the RETR was replaced with the RKKR cleavage site in both IN and NL (0.0–0.0‰ to 3.3–3.8‰ and 3.7–4.5‰, respectively; data S3).

Insertions are driven by the 1043–1048 cRNA structure

Most insertions detected in IN_{RKKR} and NL_{RKKR} were consistent with potential duplications after RdRp trapping by the 1043–1048 cRNA structure. In addition, NL_{RKKR} led to higher insertion frequencies than IN_{RKKR}, suggesting that this shared cRNA structure, which is more stable in NL, might play a more important role than the IN-specific predicted RNA structures.

To assess the role of the 1043–1048 cRNA structure in driving insertions at the HA cleavage site, mutated HAs were designed in which this structure was specifically disrupted through 7- and 2-nt substitutions in the regions flanking the IN_{RKKR} and NL_{RKKR} cleavage sites, respectively. In parallel, we designed mutated HAs with 13 or 10 substitutions, respectively, disrupting all predicted transient RNA structures. Both the targeted (Fig. 2C) and the full disruption (fig. S2B) approaches decreased heteropolymer insertion frequencies by, respectively, 59 and 66% in IN_{RKKR} (8.5–10.1‰ to 3.7–3.9‰ and 3.0–3.4‰) and by 88 and 82% in NL_{RKKR} (31.0–36.3‰ to 3.7–4.6‰ and 5.5–6.8‰). This observation suggested that the 1043–1048 cRNA structure plays an important role for heteropolymer insertions but that there is a basal heteropolymer insertion frequency in RKKR-bearing cleavage sites in the absence of predicted transient RNA structures. Complex insertion frequencies decreased by 70% for IN_{RKKR} (0.6–0.7‰ to 0.1–0.3‰) and by 87% for NL_{RKKR} (1.7–2.2‰ to 0.2–0.3‰), but single-nucleotide and homopolymer insertion frequencies showed a less consistent trend, either being stable or decreasing when predicted RNA structures were disrupted. To further investigate the inconsistent effect of structure disruption on single-nucleotide and homopolymer insertions, we designed two additional full-disruption mutants for IN_{RKKR} and two additional targeted-disruption mutants for NL_{RKKR}. Different substitutions, but with similar predicted disrupted RdRp-trapping structures, had comparable and consistent effects on heteropolymer and complex insertion frequencies (fig. S2, C and D, and data S4) but still showed inconsistent patterns for single-nucleotide and homopolymer insertions. Sorting insertions by size instead of type revealed that larger insertions, composed of heteropolymer, homopolymer, and complex insertions, were more affected by disruption approaches.

Next, we stabilized predicted transient RNA structures. We introduced three substitutions flanking the IN_{RKKR} cleavage site predicted to change the IN 1043–1048 cRNA structure to the more stable NL one. Heteropolymer insertion frequencies increased by two- to threefold (8.5–10.1‰ to 22.3–27.1‰), to levels closer to NL_{RKKR} (31.0–36.3‰; Fig. 2D). Single-nucleotide insertion frequencies showed little variation (2.4–5.1‰ to 2.5–6.7‰), but homopolymer insertions frequencies showed a twofold increase (1.0–2.5‰ to 2.3–4.7‰). Splitting insertions by size showed that 2-nt insertions increased about twofold (9.1–9.6‰ to 19.8–25.0‰), whereas larger insertions increased about threefold (1.9–2.4‰ to 5.7–8.2‰). Stabilization of only the IN-specific 1040–1042 cRNA and 1052 vRNA structures did not lead to an increase in insertion frequencies (fig. S2E).

To further corroborate the role of the 1043–1048 cRNA structure in driving heteropolymer and large insertions, we modified NL_{RKKR} in two ways predicted to stabilize this structure. First, the length of the double-stranded region was increased by two substitutions, creating two additional base pairs at the edge farthest from the RdRp. This external stabilization did not lead to an appreciable change in insertion frequencies (e.g., heteropolymer: 31.0–36.3‰ to 27.6–29.6‰; fig. S2F). The predicted 1043–1048 cRNA structure was then elongated with two additional base pairs at the edge closest to the RdRp, hypothetically tightening it around the RdRp. This internal stabilization resulted in an eightfold increase in heteropolymer insertion frequencies (31.0–36.3‰ to 275.2–285.3‰), with about 250.3 to 257.3‰ of consensus reads presenting the 1044GA heteropolymer insertion (Fig. 2D). There were also many more complex insertions (1.7–2.0‰ to 32.2–36.6‰), with 16.0 to 19.1‰ of consensus reads presenting a 1046GAGA insertion (0.2 to 0.2‰ in WT NL_{RKKR}), likely the result of two 1044GA heteropolymer insertions. Single-nucleotide insertion frequencies decreased by 77 to 80% (4.6–5.3‰ to 0.9–1.2‰), whereas homopolymer insertion frequencies remained within a twofold range (2.2–3.3‰ to 1.6–1.9‰). Both 2-nt and larger insertions increased about eightfold (28.5–33.1‰ to 252.6–259.4‰ and 6.0–8.1‰ to 49.0–57.2‰, respectively). Background insertion frequencies remained stable in non-RdRp samples of all HA structure mutants (data S3). Taken together, these results showed that, while single-nucleotide insertions might be primarily affected by the sequence in the RdRp cavity, heteropolymer and large insertions observed in H5 RKKR HAs are affected by changes in the predicted 1043–1048 cRNA RdRp-trapping structure.

As HPAIV emerge most commonly in gallinaceous birds, and to understand the potential role of host factors on insertions by the RdRp, experiments were conducted in chicken cells using NL_{RETR}, NL_{RKKR}, NL_{RKKR} 1043–1048 cRNA structure destabilization (Fig. 2C) and internal stabilization (Fig. 2D) mutants. For all templates, the observed insertion profiles—both in nature and frequency—were very similar to those seen in mammalian cells, suggesting that gallinaceous-specific host factors are probably not involved in polymerase trapping by template RNA structures and subsequent insertions (fig. S4).

We also examined the insertion profiles in two additional regions of the HA gene. There were no other regions in IN or NL HAs comprising putative transient stems enclosing A/U-rich sequences as observed at the HA cleavage site of IN_{RKKR} or NL_{RKKR}. We selected two regions in IN HA, which were the closest by substitution to AAAGAAAAAAAA (A_3GA_8), sequence present in the RdRp cavity when 1044GA heteropolymer insertions are introduced (IN_{556} and IN_{1275}, 3 and 5 nt, respectively). No insertions were observed in IN_{556} and IN_{1275} WT sequences (fig. S5). When these regions were substituted to comprise the A_3GA_8

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pattern (IN_{556} and IN_{1275} A_3GA_8), single-nucleotide insertions mainly increased, reminiscent of our previous observation that single-nucleotide insertions might be primarily affected by the sequence in the RdRp cavity. A structure analogous to 1043–1048 cRNA structure was then introduced, predicted to form around the RdRp when the G of A_3GA_8 is in the +1 position (IN_{556} and IN_{1275} 1043–1048 cRNA structure). Introduction of the structure only led to an impact on insertion frequencies in conjunction with the A_3GA_8, with insertion profiles very similar to those reported at the HA cleavage site (IN_{556} and IN_{1275} A_3GA_8 1043–1048 cRNA structure). These observations show that insertions can be detected in parts of HA other than at the cleavage site, provided the concomitant presence of RdRp-trapping stems and A/U-rich sequences in the RdRp cavity.

Heteropolymer insertions occur on cRNA

As the 1043–1048 cRNA structure was predicted to form only in cRNA sense, we hypothesized that heteropolymer insertions and insertions of >2 nt occur when a cRNA template is replicated into a vRNA product by the RdRp. We therefore developed a unidirectional RNP replication system (see methods and fig. S6). Insertion frequencies were not background-corrected, as different RNA senses were targeted in RdRp-containing and non-RdRp samples leading to differences in observed insertion frequencies. When only cRNA-to-vRNA replication occurred, similar insertion frequencies were observed compared with a bidirectional system (e.g., heteropolymer: 18.4–20.5‰ to 15.9–18.0‰; Fig. 3, A and B). In contrast, when cRNA-to-vRNA replication was blocked, heteropolymer insertion frequency was strongly decreased compared with the bidirectional system (10.6–10.9‰ to 0.7–2.2‰; Fig. 3, C and D). Additionally, frequency of complex insertion decreased >50% when cRNA-to-vRNA replication was blocked (1.8–1.9‰ to 0.2–0.7‰), although single-nucleotide (28.9–32.2‰ to 16.1–23.0‰) or homopolymer (4.7–7.7‰ to 3.1–5.4‰) insertions were not affected. Both 2-nt and larger insertions decreased by 68 and 65%, respectively (13.0–14.9‰ to 4.3–4.5‰ and 3.5–5.1‰ to 1.1–2.0‰). Taken together, these results showed that heteropolymer insertions at the IN_{RKKR} cleavage site mainly occur during cRNA-to-vRNA replication and confirmed the role of the 1043–1048 cRNA structure in driving heteropolymer and large insertions at the HA cleavage site. Of note, the 1043–1048 cRNA structure was also highly conserved in HAs of putative LPAIV precursors of HPAIV outbreaks (table S2).

Direct visualization of RdRp trapping

We obtained structural evidence for RdRp trapping by determining a high-resolution cryo–electronic microscopy (cryo-EM) structure of an actively synthesizing recombinant influenza virus RdRp stalled by a template-in/template-out stem. A model 70-mer template was designed on the basis of the NL_{RETR} 1043–1048 cRNA sequence to generate a stable trapped structure (fig. S7). We modified the template relative to the natural sequence to improve trapping and generate a homogeneous complex for cryo-EM. First, the internal stabilization mutant was used (Fig. 2D), consisting of seven base pairs, six Watson-Crick and one G:U (fig. S7, step 2). The template was then further modified so that RNA synthesis would stall when A1045 is in the +1 position, corresponding to when the 1043–1050 cRNA structure was predicted to form. This was done by substituting all A's with U's in the template 3′ of the +1 position (fig. S7, step 3) and performing unprimed RNA synthesis in the presence of adenosine triphosphate (ATP), cytidine triphosphate (CTP), guanosine triphosphate (GTP), and uridine triphosphate (UTP) that is nonhydrolyzable (UpNHpp). In this reaction, binding of UpNHpp opposite to template nucleotide A1045 at the +1 position should lead to chain termination. Next, the conserved 3′ and 5′ vRNA promoter ends were added (the cRNA promoter 3′ end contains an A), forming a functional 70-mer mvRNA (fig. S7, step 4). Of note, A34 in this 70-mer template corresponds to A1045, the expected stalling position. Full-length RNA synthesis in the presence of all nucleoside triphosphates (NTPs), despite the presence of the internally stabilized stem, and stalling in the absence of UTP were confirmed by performing radiolabeled RNA synthesis assays (fig. S8). Structure determination was attempted with this template, but no stable stem structure was observed, possibly because of the instability of the A/U-rich template-product duplex. Therefore, to avoid potential template:product interstrand slippage, four A:Us were changed to G:Cs (fig. S7, step 5). It is only with this final sequence that the stalled structure was sufficiently constrained to be observable.

Cryo-EM analysis of particles performing unprimed RNA synthesis, with UTP substituted by UpNHpp, on the 70-mer mvRNA with template-product stabilization (fig. S7, step 5) yielded two major conformations (Fig. 4A and fig. S9). The overall resolution of both structures was close to 2 Å, allowing unambiguous identification of the nucleotides in the template-product duplex within the RdRp core (Fig. 4, B and C, and fig. S9). In the first structure, a nontranslocated

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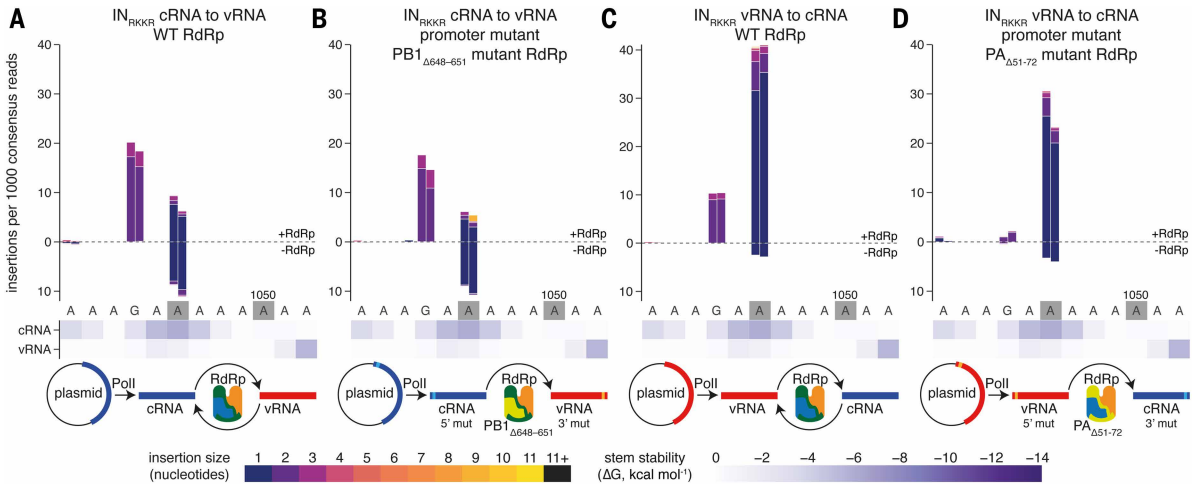

Fig. 3. Heteropolymer insertions occur during replication of cRNA templates. Insertion frequencies observed in IN_{RKKR} vRNA in a bidirectional (**A**) or unidirectional (**B**) replication system or in IN_{RKKR} cRNA in a bidirectional (**C**) or unidirectional (**D**) replication system. Using the same representation as Fig. 2. Uncorrected frequencies are shown, growing upward for RdRp-containing samples and downward for input (non-RdRp) samples. In input samples, cRNA [(A) and (B)] or vRNA [(C) and (D)] is targeted for sequencing. Schemes below the graphs show the directionality of replication in each assay.

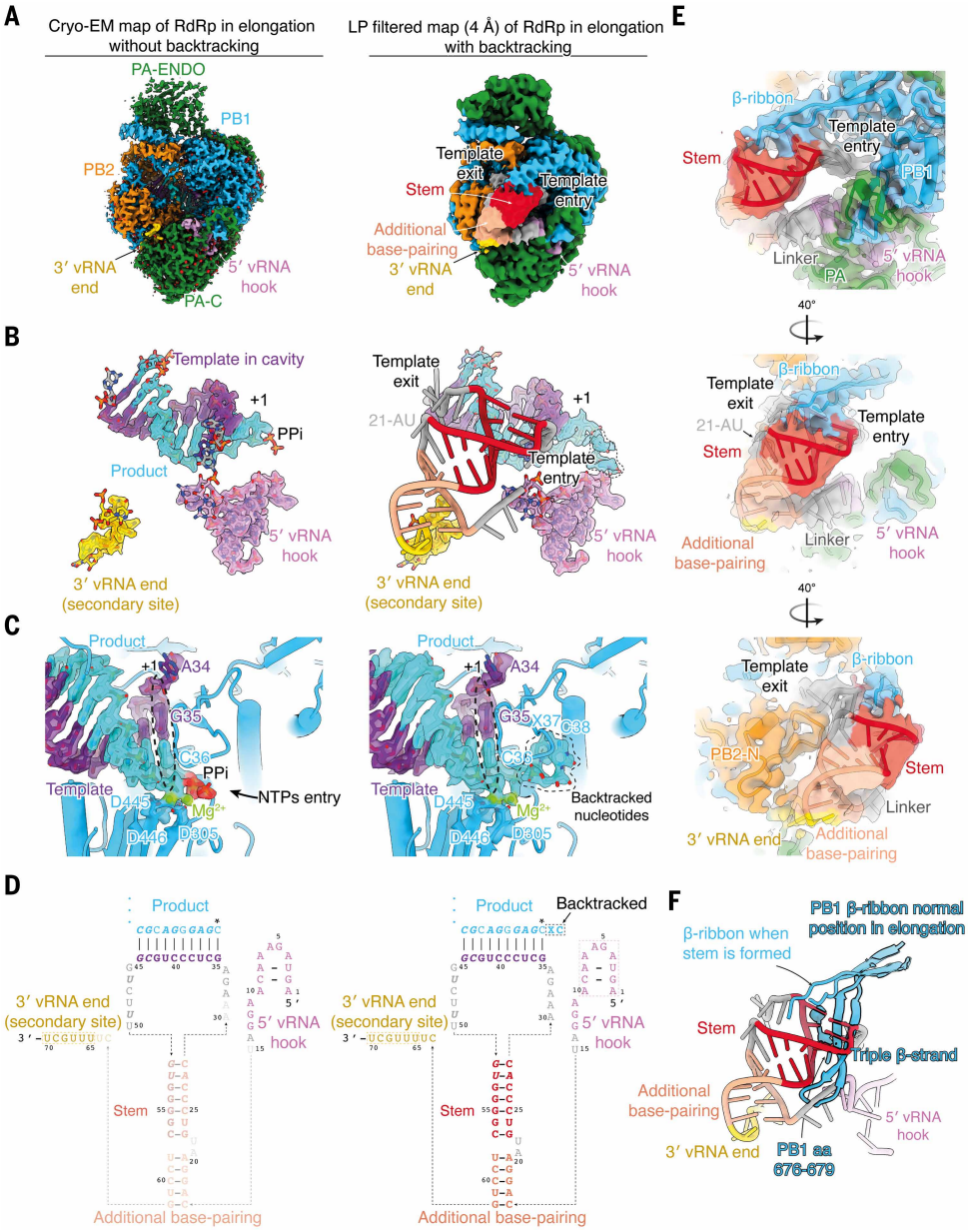

Fig. 4. Structural comparison between the two cryo-EM maps of influenza virus RdRp during elongation of the 70-mer mvRNA with template-product stabilization. (**A**) Cryo-EM maps of the RdRp without backtracking (left) or with backtracking and stem formation (right). The sharpened map is shown for the left structure. A 4-Å low-pass filter map is displayed for the right structure. The RdRp is color coded as in Fig. 1. The 5′ vRNA end, forming a hook, is in pink, and the 3′ vRNA end, bound to the secondary site (*46*), is in yellow. Water molecules in the sharpened map are colored red. The stem is colored red, additional base-pairing beige, and linker nucleotides gray. (**B**) Extracted Coulomb potential from the sharpened maps of the RdRp during elongation of the 70-mer mvRNA, without backtracking (left) or with backtracking and stem formation (right). Color coding is as in (A), with the template within the RdRp cavity in purple, and the product in light blue. The pyrophosphate (PPi) is indicated (left structure). The two backtracked pyrimidines are circled with a dashed line (right structure). The +1 position is indicated. Given its flexibility, the stem is not properly distinguishable within the sharpened map (right structure), necessitating the 4-Å low-pass filter. (**C**) Close-up view of the RdRp during elongation of the 70-mer mvRNA without backtracking (left) or with backtracking and stem formation (right). Color coding is as in (B). Base pairs in the +1 position are circled with a bold dashed line. (**D**) Schematic of the 70-mer mvRNA conformation when bound to RdRp during elongation in the structures without backtracking (left) and with backtracking and stem formation (right). The +1 position is indicated with an asterisk. Color coding is as in (B). Unresolved nucleotides are shown with transparent shading. (**E**) The stem observed within the 4-Å low-pass filtered map of the RdRp during elongation of the 70-mer mvRNA [structure corresponds to the right-hand structures in (A) to (D)]. (**F**) Comparison of PB1 β-ribbon (residues 178–213) conformation between the RdRp during elongation upon backtracking and stem formation and normal elongation (structure outlined in black, PDB ID 8PNQ). PA-ENDO, endonuclease domain of PA; PA-C, C-terminal domain of PA.

product complex was observed with template G35 at the +1 position, corresponding to one position before the expected stalling by UpNHpp (Fig. 4C, left). However, the template RNA external to the polymerase core was poorly defined. In the second structure (Fig. 4, C and D, right), two additional nucleotides were incorporated in the product followed by a 2-nt backtracking of the entire template-product duplex, leaving G35 again at the +1 position. In this case, the expected template-in/template-out stem was clearly defined outside the RdRp core, albeit at moderate resolution (~4 Å) compared with the rest of the structure (Fig. 4, A and E, and fig. S9). To circumvent the lack of nucleotide resolution for this part of the RNA, an AlphaFold3 model (*54*) was built of the double-stem region (nucleotides 16–29 and 51–62) that could be placed as a rigid body into the density (Fig. 4E) and connected by its four ends to the well-defined RNA bound to the RdRp core. The double-helical region of the stem outside the RdRp was longer than expected because it contained not only the predicted inverted helix of the internal stabilization mutant comprising 23-GUCCCAC base-pairing with 51-GUGGGGC but also an additional five–base-pair stem (16-CAGGA with 58-UCCUG) resulting from fortuitous complementarity in the 5′ and 3′ flanking regions (Fig. 4D). To accommodate the stem, the PB1 beta-ribbon was displaced compared with its normal position during elongation (*46*), leading to disruption of the triple beta-strand formed with PB1 amino acids 676–679 (which are disordered in this structure), as proposed by Pitré *et al.* (*55*) (Fig. 4F). Although the sequence of the 70-mer mvRNA template deviates considerably from the natural RNP context of replication of the HA gene, only by making these adaptations to stabilize the template-in/template-out stem could we visualize the formation of an RdRp-trapping RNA structure.

H6 cRNA structure leads to low heteropolymer insertion frequencies

HPAIV emergence has been restricted to viruses of the H5 and H7 subtypes (*1*). The reasons for this subtype restriction are still unknown, although we and others have suggested that it might be due to codon use at the HA cleavage site favoring substitutions to insertion-prone sequences in H5 and H7 HAs (*17*, *20*, *22*, *39*, *42*). To assess the potential impact of transient RdRp-trapping structures in non-H5/H7 HAs, we chose to study H6. H6 HAs are phylogenetically closely related

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to H5 HAs, and H6 LPAIVs circulate extensively in poultry (*56*). The A/mallard/Sweden/81/2002 HA (SW) was used, as its cleavage site amino acid sequence QIETR corresponds to the H6 LPAIV nucleotide consensus. Additionally, SW presents a predicted RdRp-trapping cRNA structure spanning +1 positions 1038 to 1043, equivalent to positions 1043 to 1048 in H5 HAs (H5 numbering will be used hereafter). This structure is almost identical to the NL 1043–1048 cRNA structure, but with an A-U instead of a G-C base pair at the third position. RNPs consisting of the H5 A/Indonesia/5/2005 RdRp and NP and the H6 SW_{QIETR} RNA were produced, and only single-nucleotide insertions at extremely low frequencies were detected (−0.0006 to 0.0‰; Fig. 5A), similar to IN_{RETR} (−0.007 to 0.002‰) and NL_{RETR} (−0.09 to −0.005‰). After replacing the H6 QIETR cleavage site with the QRKKR of NL_{RKKR} (equivalent to substituting to A_3GA_8 at the nucleotide sequence), heteropolymer insertion frequency increased to 4.2–5.0‰ (Fig. 5B), levels lower than in IN_{RKKR} and NL_{RKKR} but comparable to the basal heteropolymer insertion frequencies in RKKR-containing H5 HAs with disrupted structures (ranging between 1.8 and 6.8‰). Accordingly, disrupting the predicted H6 1043–1048 cRNA structure by introducing two substitutions did not reduce insertion frequencies, with all insertion types varying less than twofold (e.g., heteropolymer frequencies 4.2–5.0‰ to 4.0–5.4‰; Fig. 5C). However, stabilization of the predicted H6 1043–1048 cRNA structure, by introducing the NL G-C base pair in the third position of the structure, increased heteropolymer insertions about fivefold (4.2–5.0‰ to 20.7–20.9‰; Fig. 5D), levels closer to those observed in NL_{RKKR} (31.0–36.3‰). As observed previously, single-nucleotide frequencies showed little variation (4.2–4.7‰ to 4.5–8.2‰), whereas homopolymer insertion frequencies increased twofold (1.4–1.5‰ to 2.6–3.3‰). The increase was more pronounced for large insertions, with 2-nt insertions increasing fourfold (5.3–6.2‰ to 21.4–21.9‰) and larger insertions increasing sixfold (0.5–0.5‰ to 2.5–3.1‰). These results suggest that, in addition to the codon use at the cleavage site, predicted RdRp-trapping structures play an important role in the potential for HPAIV-yielding insertions in non-H5 HAs.

Discussion

The mechanisms underlying recurrent insertions at the H5 HA cleavage site have remained unsolved for decades. The importance of the HA cleavage site sequence has become increasingly supported (*22*, *25*, *32*–*39*), and here we have provided empirical evidence of the involvement of transient RNA structures. We show that the trapping of the influenza virus RdRp on A/U-rich sequences by transient RNA structures formed by the template is the mechanism by which nucleotide insertions occur at the H5 cleavage site. The involvement of the transient RNA structures was shown by model-guided stabilization or disruption of the 1043–1048 cRNA structure, leading to the expected impact on insertion frequencies, and by direct visualization of template RNA stem structures forming outside the RdRp by structural analyses. In the cryo-EM structures, the overall template trajectory was fully consistent with models of elongation proposed previously in the context of transcription (*46*) and of proposed t-loop formation in the context of mvRNA replication and transcription (*40*, *55*). The observation of 2-nt backtracking provided direct insight into the mechanism of nucleotide insertion. In the structure where the stem was visible, concerted backtracking of both template and product was observed, which was probably a result of the stabilization of the template-product duplex, preventing its disruption through interstrand slippage. When the native 1043–1048 cRNA structure is formed, the template-duplex corresponds to the HA cleavage site sequence. With a weak A/U-rich template-product duplex, as in NL_{RKKR}, interstrand slippage could occur as in polyadenylation (*46*), leading to the sole backtracking of the template, with the product staying in place. A 2-nt backtracking of only the template would result in a 2-nt insertion in the product, which was the most frequent event observed in our vRNP reconstitution assay in cells. Together, these data explain how the combined effect of RdRp-trapping structures in concert with a weak template-product duplex leads to insertions and why A/U-rich cleavage site sequences have been identified as insertion-prone (*22*, *25*, *32*–*39*). Similarly, an A or U stretch in the template RNA is also required for mRNA polyadenylation (*57*). Using unidirectional replication systems, we also showed that heteropolymer insertions were mainly generated during cRNA-to-vRNA replication, corroborating the important role of the 1043–1048 cRNA structure. Results from the unidirectional assays also confirmed that heteropolymer insertions at the H5 cleavage site indeed occur on templates containing purine rather than pyrimidine stretches, as has been commonly assumed (*8*, *22*, *25*). As G:U base pairs can form in RNA molecules, A stretches with interspersed Gs might be more favorable for realignment than A stretches interrupted by any other nucleotide, as suggested previously (*8*). Deletions at the HA cleavage site were sensitive to A/U richness but showed very little sensitivity to transient RNA structure changes relative to insertions (data S4 and S7). Similar insertion profiles were observed in mammalian and chicken cells, suggesting that species-specific host factors are likely not involved in this process. These results reinforce previous findings (*58*, *59*), highlighting interspecies differences in intrahost selection as key determinant of the species-specific emergence of HPAIV. A detailed animation showing how the most commonly

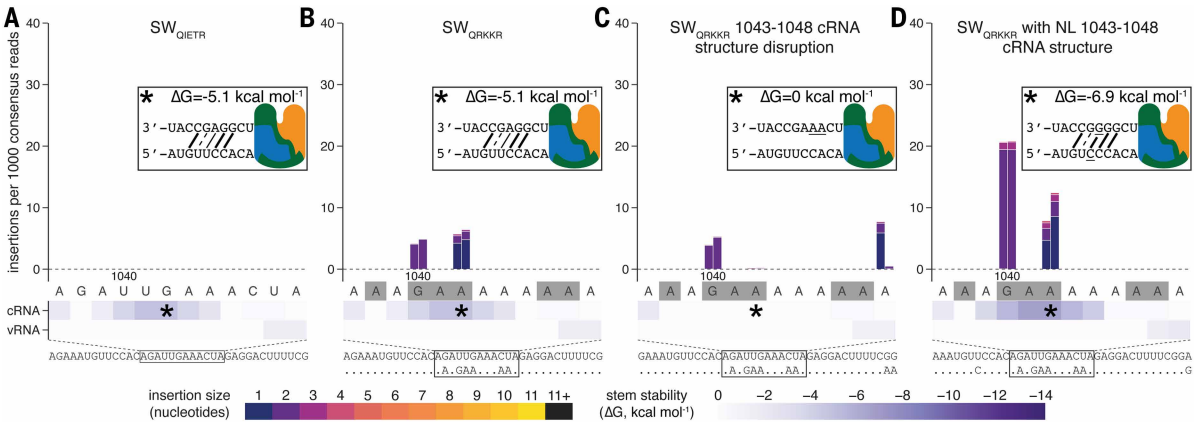

Fig. 5. The cRNA structure of H6 A/mallard/Sweden/81/2002 HA (SW) induces less heteropolymer insertions than the H5 NL 1043–1048 cRNA structure. Insertion frequencies observed in the consensus H6 QIETR cleavage site (**A**), with a QRKKR cleavage site (**B**), with a QRKKR cleavage site and substitutions disrupting the predicted 1043–1048 (H5 equivalent position) cRNA structure (**C**), or with a QRKKR cleavage site and substitutions stabilizing the predicted 1043–1048 cRNA structure (**D**). Using the same representation as Fig. 2. Uncorrected insertion frequencies for RdRp-containing and background samples are provided as an interactive figure in data S3 or available online (*47*).

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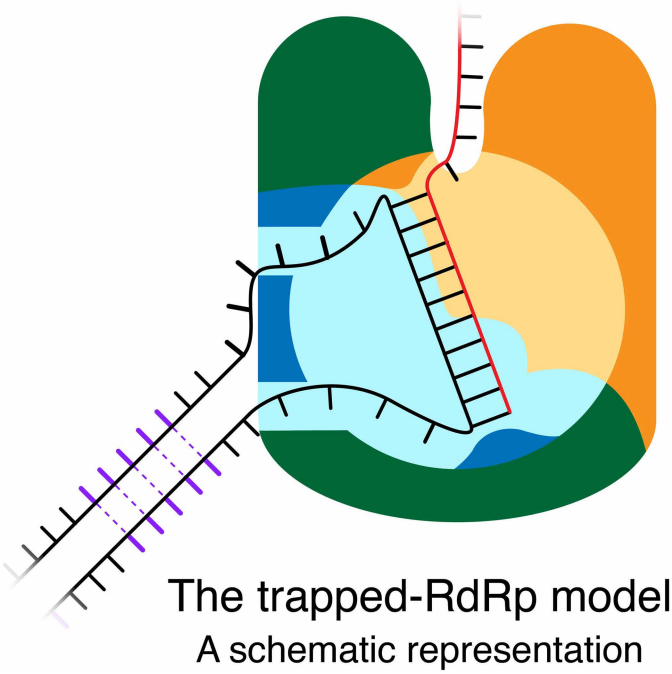
observed insertion, 1044GA, could be explained by the trapped-RdRp model is provided in Movie 1.

Here, we observed that insertions at the H5 HA cleavage site are rare unless substitutions are introduced that change the LPAIV RETR cleavage into a more insertion-prone RKKR sequence. The frequency of in-frame insertions that might lead to functional HA proteins remained very low, in line with the rarity of HPAIV conversions in nature. The in-frame insertions we observed led to cleavage sites that were similar or identical to H5 HPAIV MBCS sequences observed in nature (RRKKR, RKRKKR, RRRKKR). In fact, starting from a substituted cleavage site sequence (REKR, RKTR, or RKKR), 10 of the 11 MBCSs with insertion of known H5 HPAIVs can be explained by insertions observed in our system. These 10 MBCSs require either a single heteropolymer occurrence for six of them, or several consecutive heteropolymer or combinations of hetero- and homopolymer insertions for another four. If MBCSs can result from several consecutive insertion events occurring on the same molecule, frameshifting insertions could also play a role in MBCS acquisition. Although the need for several insertions on the same molecule to restore the frame reduces the odds of occurrence, additional insertions may have higher chances of accumulating in templates already bearing insertions, leading to a more repetitive nucleotide cleavage site sequence. Only the MBCS of the H5 HPAIV responsible for the 1961 outbreak in South Africa seems unlikely to have occurred solely by duplication and may be the result of NHR (*1*).

Along with providing a mechanism for insertions at the H5 HA cleavage site, our study provided insights regarding the subtype specificity of HPAIV emergence. Although we focused on insertions in H5 cleavage sites, H7 HPAIV MBCS acquisitions commonly occurred after duplication-like insertions. RNA sequences and structures found at H7 LPAIV cleavage sites differ markedly from those found in H5 LPAIVs (*21*, *42*), yet a similar structure-driven mechanism might be involved. Introducing the H5 QRKKR sequence into an H6 HA, leading to a weaker template-product duplex, only led to a low frequency of insertions, probably reflecting basal insertion levels in an A-rich sequence.



Movie 1. The trapped-RdRp model. A detailed animation showing how the most commonly observed insertion, 1044GA (cRNA sense), is explained by the trapped-RdRp model. Using the same representation as Fig. 1, RNA structures at the bottom are in dot-bracket format: {} for GC, () for AU, <> for GU base pairs.

Replacing the weaker H6 cRNA structure with the H5 cRNA structure increased insertions to ranges more comparable to those in H5 RKKR cleavage sites. Once again, this highlights the synergism between strong transient RNA structures blocking the polymerase on a weak template-product duplex and indicates that subtype specificity of HPAIV emergence might be explained by the combined presence of these factors. In a previous study, we showed that fewer substitutions were necessary in H5 and H7 HAs than in HAs from other subtypes to acquire A-rich cleavage site sequences, owing to differences in codon use (*42*). Whereas only two substitutions were required to convert the H5 LPAIV cleavage site to QRKKR, six substitutions were needed in H6, making such a change far less likely to occur naturally. Notably, about 3% of H6 HAs have the full NL 1043–1048 cRNA structure, indicating that H6 HPAIV emergence may primarily be prevented by codon use at the cleavage site, disadvantaging acquisition of an A-rich sequence (*42*).

We acknowledge three limitations. First, the minimal requirements for transient structures to impede influenza RdRp replication have not been refined. The effect of transient structures may depend not only on quantitative parameters, such as structure stability or RNA folding kinetics, but also on qualitative parameters, such as structure tightness around the RdRp as well as the nature of the base pair forming closest to the RdRp. The position of the RNA structure in relation to the sequence present in the RdRp active site also plays a crucial role, as illustrated by the fact that stabilizing the 1043–1048 cRNA structure increased insertions, whereas stabilizing the IN-specific 1040–1042 cRNA structure did not. Second, we have not investigated whether the NP-binding profile of viral RNAs might affect the formation of RdRp-trapping structures. Recent work has shown that NP-binding profiles differ depending on the genomic sequence of the influenza virus involved (*28*, *60*). NP-bound RNA is thought not to be able to form secondary structures (*61*). However, during replication, NP molecules are dislodged from the template (*46*). Recent structures of vRNPs during transcription suggested that at least two NP molecules might be simultaneously dislodged to allow the template to enter the polymerase (*62*), which might facilitate formation of transient RNA structures between the (transiently) naked RNA strands protruding from the template entry and exit channels. Third, experiments in this study were conducted using only one influenza virus polymerase, that of A/Indonesia/5/2005, and it cannot be excluded that properties of the RdRp itself might also influence the sensitivity to RdRp-trapping structures (*40*, *63*, *64*).

As detailed above, the mechanism of insertions at the HA cleavage site we have discovered is reminiscent of those proposed in very different influenza virus contexts, such as mRNA polyadenylation (*46*) and mvRNA replication and transcription (*40*, *55*). RdRp replication errors induced by RNA structures might therefore be a more widespread phenomenon with different effects. For example, this mechanism may be involved in the recurrent insertions and deletions observed in the 160 antigenic loop of human influenza B viruses (*65*, *66*). Beyond influenza viruses, transient RdRp-trapping structures may also affect the replication of other negative-sense RNA viruses with similar polymerase entry and exit channel architecture, such as Ebola virus, Lassa virus, measles virus, or rabies virus (*67*).

Materials and methods

Prediction of RdRp-trapping structures

Putative RdRp-trapping structures were predicted by a Python script using a sliding window approach and minimal free energy RNA structure prediction with the ViennaRNA (*68*) package. A schematic representation of the method is shown in fig. S1. Nucleotides within an identically sized window up- and downstream of the RdRp footprint, corresponding to the exiting and entering template strands, respectively, were considered for sequence prediction. To mimic the close proximity of the entering 5′ and exiting 3′ template strand ends, the nucleotides in the RdRp footprint were replaced by an arbitrary sequence (AAGGGGGGAAAACCCCCCAA), constrained to fold as a stem-loop [constraint: xx((((((xxxx))))))xx, where x denotes nucleotides forbidden to pair, while open parentheses are forced

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to pair with the corresponding closed parentheses]. Folding was then performed using the vrna_mfe function while disallowing lonely base pairs. After folding, the minimum free energy value was corrected by the minimum free energy value obtained with the arbitrary stem alone. The resulting standard Gibbs free energy change at 37°C ($\Delta G°_{37}$) is denoted as ΔG. The script allows for variation in the size of the RdRp footprint (20 for figures in the main text), the position of the active site from the 5′ end of the sequence within the RdRp footprint (five for figures in the main text), and the size of the flanking sequence window to consider. A 10-nt window was used throughout the main text as, first, transient structures that would affect replication are expected to form close to the RdRp. Second, parts of the genome are bound by NP, which is only temporarily dislodged by the RdRp during replication. Prediction windows encompassing larger flanking regions have a higher risk of including NP-bound nucleotides, which would not be available to form base pairs. The interactive data S1 shows the RNA structures predicted with a 10- or 15-nt window for all HA sequences used. Code is available on Zenodo (*69*) or on Github (*70*).

Plasmids

The generation of pCAGGS expression plasmids for A/Indonesia/5/05 PB2, PB1, PA, and NP was described previously (*71*). Plasmids expressing the mutated PB1 and PA genes were obtained by primer-directed polymerase chain reaction (PCR) mutagenesis, as described previously (*72*). The generation of modified pHW2000 plasmids expressing genomic vRNAs of the HA segment of A/mallard/Netherlands/3/1999 (NL, NCBI:txid680899), A/mallard/Sweden/81/2002 (SW, NCBI:txid753608), and A/Indonesia/5/2005 (IN, NCBI:txid400788), whose MBCS was replaced by the LPAIV RETR cleavage site, under the control of the human RNA polymerase I promoter were described previously (*73*–*75*). Mutated HA templates were subsequently generated by primer-directed PCR mutagenesis (*72*). For the unidirectional RdRp replication systems, the IN_{RKKR} segment was transferred, in both vRNA and cRNA sense, to a pSP72 vector under control of the human RNA polymerase I promoter (pSP72-PhuTmu) (*73*) by "megaprimer" PCR mutagenesis, as described previously (*76*). Influenza promoter substitutions were then introduced into these plasmids through primer-directed PCR mutagenesis (*72*). For the vRNP reconstitutions assays in chicken cells, the human Pol I promoter was replaced with the chicken Pol I promoter. Briefly, the human Pol I promoter was removed from pSP72-PhuTmu by deletion PCR. The chicken Pol I promoter (GenBank accession no. DQ112354, nucleotides 590–1014) was then PCR-amplified with primers introducing AarI restriction sites at both ends and inserted between BbsI sites. NL_{RETR} HA, NL_{RKKR} HA, or NL_{RKKR} HA stabilization or destabilization structure mutants were subsequently cloned downstream of the chicken Pol I promoter in vRNA orientation using In-Fusion Cloning (Takara), following the manufacturer's instructions. Primers were ordered from Eurogentec. All plasmids were sequenced using a 3130XL or 3500XL Genetic Analyzer (Applied Biosystems) to confirm introduced mutations.

Cells and generation of influenza virus vRNPs

Human embryonic kidney cells (293T, ATCC) and chicken embryonic fibroblast cells (DF-1, ATCC) were cultured in Dulbecco's modified Eagle's medium (Capricorn) supplemented with 10% fetal bovine serum (FBS, Sigma), 100 IU/ml penicillin, 100 μg/ml streptomycin, 2 mM glutamine, 1 mM sodium pyruvate, nonessential amino acids. 293T cells were further supplemented with 500 μg/ml of geneticin (Gibco). 293T cells were transfected using $CaCl_2$, as described previously (*73*). For DF-1 transfections, cells were plated 1 day prior at a density yielding ~80% confluence at the time of transfection, maintained in antibiotic-free medium, and transfected using TransIT-X2 reagent (Mirus Bio) according to manufacturers' instruction with minor modifications. Briefly, 5.6 ug of total plasmid DNA was diluted in 1.4 ml Opti-MEM I Reduced-Serum Medium (Invitrogen) and mixed with 22.4 μl TransIT-X2 reagent. After 15 to 30 min of incubation at room temperature to allow complex formation, the mixture was added dropwise to cells. Media was replaced 5 hours after transfection. Cells were then incubated for 48 hours before harvesting.

RNPs were generated in two biological replicates using the A/Indonesia/5/2005 RdRp (PB2, PB1, PA) and NP and an HA vRNA or cRNA as template. To this end, 293T cells, cultured in 10-cm dishes, were transfected using 2.5 μg of plasmids each encoding a A/Indonesia/5/2005 RdRp subunit (PB2, PB1, PA) mRNAs, 10 μg of plasmid encoding A/Indonesia/5/2005 NP mRNA, 5 μg of plasmid expressing HA cRNA or vRNA templates and 7.5 μg of empty pCAGGS plasmid. For DF-1 cells, the same plasmid ratios were used, but with adjusted amounts: 0.6 μg of plasmids each encoding an RdRp subunit (PB2, PB1, PA) mRNA, 2.4 μg of plasmid encoding NP mRNA, 1.2 ug of plasmid expressing HA cRNA or vRNA template, and 0.2 μg of empty pcDNA3 plasmid. In parallel, background insertion levels, expected to be primarily due to errors by the human Pol I during generation of the initial RNA templates from plasmid DNA, were quantified in two biological replicates. For background samples, the RdRp-encoding plasmids were omitted, and an additional 7.5 μg of empty pCAGGS plasmid for 293T or 1.8 μg of empty pcDNA3 plasmid for DF1 cells was added. vRNP reconstitutions assays were performed in BSL-2 conditions.

Circular sequencing

To reliably identify rare insertions, we used a custom circular sequencing method (*41*). RNA molecules are circularized, leading to cDNA tandem repeats upon reverse transcription. After sequencing, repeats are collapsed into consensuses, and only consensus mutations are kept, as those are most likely present in the original RNA molecule. To evaluate the variation between independent experiments, IN_{RKKR} was included as an internal control in most rounds of RNP generation and circular sequencing, resulting in 21 independent biological replicates (fig. S3A). The two replicates with the most consensus reads were used as representatives in the main text.

RNA extraction, fragmentation, and circularization: RNAs were extracted ~48 hours after RNP transfection in 293T or DF-1 cells using the High Pure RNA Isolation Kit (Roche) according to the manufacturer's instructions. DNA was removed by performing 30-min digestion with TURBO DNAse with subsequent bead clean up (Invitrogen) according to manufacturer's instructions. Fifteen to thirty microliters of RNA, depending on cell density, was concentrated to 7.5 μl using the Oligo Clean & Concentrator Kit (Zymo Research) and fragmented by alkaline hydrolysis by adding 1 volume of 100 mM bicarbonate (Sigma-Aldrich) pH 9.2 buffer and incubating at 95°C for 10 min. After adding 1 volume of Gel Loading Buffer II (Invitrogen), RNA was run on a prewarmed denaturing 8 M urea (Sigma-Aldrich) 12% polyacrylamide (BioRad) gel in 1x Tris-Borate-EDTA running buffer (Sigma-Aldrich) at 200 V for 2.5 hours. Gels were stained with SYBR Safe (Invitrogen), diluted 1/10000 in running buffer, for 15 min under shaking and protected from light. Fragmented RNA smears were visualized by blue-light transillumination, and slices corresponding to the 60-to-120-nt region were cut and placed into a 0.5 ml microcentrifuge tube (Eppendorf), in which a hole was punctured at the bottom. These tubes were placed into 2 ml microcentrifuge tubes (Eppendorf) and centrifuged at 17,000*g* for 2 min to crush slices into a slurry. RNA was extracted, precipitated, and circularized, as previously described (*41*), with the addition of 20% v/v dimethyl sulfoxide (Sigma-Aldrich) and 10% v/v PEG 8000 (New England Biolabs) during ligation.

Reverse transcription and second strand synthesis: Ligation reactions were cleaned up with the Oligo Clean & Concentrator Kit and reverse-transcribed using Superscript IV RT (Invitrogen) according to manufacturer's instructions, using 1 μl of a gene-specific primer (10 μM; sequences presented in table S1) bearing the M13 forward (−21) sequence followed by a unique molecular identifier (UMI; to distinguish unique molecules from PCR duplicates in the downstream analysis) of 10 random nucleotides as 5′ tag (5′-TGTAAAACGACGGCCAGTNNNNNNNNNN), and extending the 55°C incubation step to 1 hour.

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After reverse transcription (RT), cDNAs were purified using 0.65 volumes of Ampure XP beads (Beckman) according to manufacturer's instructions to select for large cDNAs containing several repeats. cDNAs were eluted off the beads using 20 µl of water, and second-strand synthesis reactions were assembled by adding 3 µl of a gene-specific primer (2 µM; sequences presented in table S1) with the T7 promoter sequence as 5′ tag (5′-TAATACGACTCACTATAGGG), 3 µl of 10x PfuUltra II Fusion buffer, 0.6 µl of PfuUltra II Fusion HS DNA polymerase (Agilent), 0.75 µl of deoxynucleoside triphosphate (dNTP; 10 mM each, Roche) and 2.65 µl of nuclease-free water. Reactions were incubated at 95°C for 5 min, 55°C for 1 min, and 72°C for 15 min, purified using 0.65 volumes of Ampure XP beads and eluted in 30 µl of water.

PCR amplification and Illumina library preparation: Eluted double-stranded DNA (dsDNA) was amplified in three parallel PCR reactions using AmpliTaq Gold DNA polymerase (Applied Biosystems) with 5 µl of T7 promoter sequence primer (2 µM), 5 µl of M13 forward (−21) primer (2 µM), 5 µl of AmpliTaq $MgCl_2$ solution, 5 µl of 10x AmpliTaq Gold buffer, 1 µl of AmpliTaq polymerase, 1.25 µl of dNTP (10 mM each), and 10 µl of purified dsDNA in a total reaction volume of 50 µl. Reactions were incubated at 95°C for 5 min, followed by 30 cycles of 95°C for 1 min, 50°C for 30 s, 72°C for 2.5 min, and a final step at 72°C for 6 min. PCR reactions were loaded on 1% agarose gels, the 300 to 500 bp region was excised and DNA was extracted using the MinElute Gel Extraction Kit (Qiagen).

DNA concentrations were measured using the Qubit dsDNA HS kit (Invitrogen), and 100 ng was used for paired-end Illumina sequencing library preparation using the KAPA HyperPlus Library Preparation Kit (Roche) and the KAPA Unique Dual-Indexed Adapter Kit (Roche), according to manufacturer's instructions, starting directly with the end repair and A-tailing step, replacing the double-sided selection step by a second Ampure Bead clean up step using 1 volume of beads, extending the 72°C step during the PCR library amplification to 45 s, and performing the last clean up step with 0.7 volumes of beads. Concentrations of libraries were determined using the Qubit dsDNA HS kit, and libraries were pooled equimolarly and sequenced using the Illumina V3 Miseq (2 × 300 cycles) or Illumina Nextseq 2000 (2 × 300 cycles) platform.

Determining nature and frequency of insertions and deletions: Paired FASTQ files from Illumina paired-end sequencing were merged using AdapterRemoval (*77*) (v. 2.3.2), and consensuses were extracted from repeat-containing reads and aligned to the reference by Bowtie2 (*78*) using the CirSeq software (*79*, *80*), adapted to keep consensuses with insertions or deletions (indels) instead of discarding them. Because of low-frequency contamination of some transfections with A/Indonesia/5/2005 MBCS HA, leading to the presence of this sequence in some samples, including samples without RdRp, all consensuses bearing the A/Indonesia/5/2005 MBCS were excluded from further analysis (at most 0.04% of consensuses). Given the size of the MBCS (24 nt instead of 12 nt) and the presence of a C in the A/Indonesia/5/05 MBCS stretch, these consensuses could be reliably separated from other consensuses bearing duplication-like insertions. SAM files containing the aligned consensuses were then processed using a Python script identifying consensuses with indels and calculating the coverage at each position; Python script is available at Zenodo (*69*) and Github (*70*). Consensuses containing more than four separate indel events or a 1-nt insertion and a 1-nt deletion occurring within 5 nt from each other were considered to result from contamination with a closely related HA template and discarded. To account for duplication of identical cDNAs by PCR, the UMI (10-nt N stretch in the RT primer, see above) associated with each consensus was identified using the edlib alignment library (*81*), and consensuses with the same UMI and alignment to the reference (as determined by alignment start position and CIGAR string) were filtered out to keep a single representative. Consensuses for which the UMI could not be identified were discarded.

Insertions in repetitive sequences can often be aligned to the reference in several valid ways. Aligning these insertions consistently requires an arbitrary rule for standardizing insertions placement. Noncomplex ambiguously positioned indels were identified and realigned by deducing every possible alignment for a given indel, calculating the stability of predicted RdRp-trapping structures at each possible position (on cRNA and vRNA, using the same approach as above) and realigning the indel to the position corresponding to the lowest free energy structure. For insertions, only alignments consistent with sequence duplication were kept. In addition to standardizing insertion placement, this allowed us to test compatibility with our model by identifying putative RdRp-trapping structures that might be involved in insertions. In HA templates with disrupted structures, if no structure was predicted in any possible position, the structures of the corresponding WT HA were used instead to facilitate comparison. Unambiguous indels were not realigned as, by definition, they have a single possible alignment. Complex indels were not realigned as they probably result from several independent indel events potentially occurring at several different positions. After prediction-based realignment, frequencies for each insertion were calculated on the basis of the coverage at each given position. Insertion frequencies were then corrected by subtracting the background frequencies observed without RdRp, with insertions realigned the same way, to allow direct comparison between repeats. All raw uncorrected insertion frequencies for RdRp-containing and background samples are provided as an interactive figure in data S3, alongside additional information such as alignments, predicted structures, and coverage at each position.

Interactive HTML files were generated using Plotly (*82*) and Jinja2 (*83*). Numbers and figures in the text reflect insertion and deletion frequencies between positions 1041 and 1052 (or equivalent in H6), as they are the most relevant for MBCS acquisition.

Analysis of cleavage-site protein sequences

SAM files containing aligned consensuses were screened for insertions or deletions. Detection of contaminants and PCR duplicates was performed as above. If the difference between the number of inserted and deleted nucleotides was not a multiple of three, the cleavage site was considered to be frame-shifted. If the result was a multiple of three, the insertions and deletions were introduced into the corresponding reference sequence, and the resulting sequence was translated to obtain the protein sequence. Potential substitutions were not taken into account. To determine whether insertions and deletions occurred in the HA cleavage site region, it was defined here as the translation of nucleotides between positions 1037 and 1057 for H5 references and 1032 and 1052 for H6 references. Frequencies were calculated by dividing counts by the maximum coverage observed in the RdRp-containing sample of each replicate and were not background-corrected. The interactive HTML file was generated using Plotly (*82*) and Jinja2 (*83*).

Unidirectional RdRp replication systems

Unidirectional RdRp-replication systems were established by RNP transfection, as described above. To allow only vRNA to cRNA replication, a cRNA 3′ promoter deficient in the internal initiation step necessary for vRNA production (C1763A, cRNA sense) (*84*) was used in combination with the A/Indonesia/5/2005 RdRp carrying a PA with the 51–72 loop replaced by a GGS linker ($PA_{\Delta 51–72}$) (*85*), rendering the RdRp deficient in vRNA production. To allow only cRNA-to-vRNA replication, a vRNA 3′ promoter deficient in the terminal initiation step required for cRNA production (G2C and C9G, cRNA sense) (*84*) was used in combination with the A/Indonesia/5/2005 RdRp carrying a PB1 priming loop deletion ($PB1_{\Delta 648–651}$), which renders the RdRp deficient in vRNA production (*86*). Owing to unidirectional replication, different RNA senses had to be targeted for sequencing in RdRp-containing and non-RdRp (input) samples. In the vRNA-to-cRNA replication system, negative-sense RNA (vRNA template, Pol I product) was targeted in the input samples, and positive-sense RNA (cRNA and mRNA, RdRp product) was targeted in the RdRp-containing samples. Opposite-sense RNAs were targeted in the cRNA-to-vRNA replication system.

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Primer extension assay

Primer extension was performed as described previously (*87*). In brief, primers specific for 18*S* rRNA (5′-GAGCCATTCGCAGTTTCACTGTAC), IN_{RKKR} vRNA (5′-CCTAGCACTGGCAATCATGATGG), or IN_{RKKR} cRNA/mRNA (5′-CATGATTGTGTCAACCTGCTCTG) were radioactively labeled using T4 polynucleotide kinase (New England Biolabs) and γ-32P-ATP (PerkinElmer). RT was performed on 2.5 μl of total RNA used for CirSeq processing (see above) using the radiolabeled primers and Superscript III reverse transcriptase (Invitrogen), resulting in labeled cDNAs of 96, 126, or 136 nucleotides for 18*S* rRNA, HA vRNA, or HA cRNA, respectively. HA mRNAs are amplified by the cRNA primer but run more slowly owing to the presence of about 10 to 13 nt added through cap-snatching alongside the 5′ cap (*88*). RT reactions were run on a denaturing urea polyacrylamide gel, which was imaged by overnight exposition of a BAS-IP MS 2040 E phosphor screen (Cytiva) scanned using a Typhoon FLA 9500 (GE Healthcare).

Cryo-EM experiments

Cloning, expression, and purification of influenza A/Zhejiang/DTID-ZJU01/2013 (H7N9) RdRp: The previously described, pFastBac Dual vector encoding the influenza A/Zhejiang/DTID-ZJU01/2013 (H7N9) polymerase (Polymerase Acidic protein "PA," Uniprot: M9TI86; Polymerase Basic protein 1 "PB1," Uniprot: M9TLW3; Polymerase Basic protein 2 "PB2," Uniprot: X5F427) 4M monomeric mutant bearing the mutations PA/E349K, R490I, PB1/K577G, and PB2/G74R, with only the PB2 C-terminal Twin-strep-tag, was used as a starting point (*89*). Additional $PB2_{E627K}$ and $PB2_{N701D}$ mutations were introduced by a combination of PCR and Gibson assembly. Plasmid sequence was confirmed by Sanger sequencing for each polymerase subunit.

Influenza A/Zhejiang/DTID-ZJU01/2013 (H7N9) RdRp was expressed and purified as previously described (*89*). Briefly, the RdRp was expressed in *Trichoplusia ni* High Five cells. Cells were disrupted by sonication in lysis buffer (50 mM HEPES pH 8, 500 mM NaCl, 2 mM TCEP, 5% glycerol) with cOmplete EDTA-free Protease Inhibitor Cocktail (Roche). After lysate centrifugation, ammonium sulfate precipitation was performed. The resuspended proteins in lysis buffer were subjected to Strep-Tactin affinity purification (IBA, Superflow), and bound proteins were eluted using lysis buffer supplemented with 2.5 mM *d*-desthiobiotin. Protein-containing fractions were pooled and diluted in 50 mM HEPES pH 8, 250 mM NaCl, 2 mM TCEP, and 5% glycerol before loading onto a HiTrap Heparin HP 5 ml affinity column (Cytiva). Bound proteins were eluted using a continuous salt gradient from 250 mM to 1 M NaCl. Pure and nucleic acid-free FluPolA was dialyzed in final buffer (50 mM HEPES pH 8, 500 mM NaCl, 2 mM TCEP, 5% glycerol), concentrated, flash-frozen, and stored at −80°C for later use.

Expression of host factors interacting with the RdRp: The N-terminal His-tagged human ANP32A (huANP32A) was expressed and purified as previously described (*90*). Briefly, huANP32A was expressed in BL21(DE3) *Escherichia coli* cells through IPTG induction. Cells were disrupted by sonication in lysis buffer [50 mM HEPES pH 8, 150 mM NaCl, 5 mM β-mercaptoethanol (BME)] with cOmplete EDTA-free Protease Inhibitor Cocktail. After lysate centrifugation, the soluble fraction was loaded onto a HisTrap HP 5 ml column (Cytiva). Bound proteins were washed with 50 mM imidazole and eluted with 500 mM imidazole in lysis buffer. The N-terminal His-tag from huANP32A was cleaved using N-terminal His-tagged TEV protease. Tag-cleaved huANP32A was subjected to a Ni-sepharose affinity column to remove the TEV protease. The resulting huANP32A was concentrated and subjected to size-exclusion chromatography (SEC) using a Superdex 200 Increase 10/300 GL column (Cytiva) in a final buffer containing 50 mM HEPES pH 8, 150 mM NaCl, and 2 mM TCEP. Fractions containing huANP32A were concentrated, flash-frozen, and stored at −80°C for later use. The RNA polymerase II C-terminal domain peptide, consisting of six repeats with phosphorylated S5 6x(YSPTS*PS) (pS5 CTD), was synthesized by Cambridge Peptides.

RdRp radioactivity RNA synthesis assays: For radioactivity RNA synthesis assays, and depending on the reaction composition presented in fig. S8, 0.8 μM H7N9 RdRp was mixed with 0.8 μM 5′ vRNA 1-14 (5′ -pAGU AGA AAC AAG GC – 3′), 0.8 μM 70-mer mvRNA without template: product stabilization (5′- pAGU AGA AAC AAG GAU CAG GAA UGU CCC ACA AAG AGU UUC UUG UGG UCU UUG UGG GGC UCC UGC UUU UGC U -3′), 8 μM huANP32A, 16 μM pS5 CTD (6 repeats) (respective molar ratio: 1 RdRp: 1 template: 10 huANP32A: 20 pS5 CTD). Reactions were initiated at 30°C for 4 hours by adding ATP/GTP/CTP or all NTPs, α-^{32}P ATP (PerkinElmer), pppApG (Jena Biosciences), and $MgCl_2$ in a final assay buffer containing 50 mM HEPES pH 8, 150 mM NaCl, 2 mM TCEP, 100 μM/NTP, 1 mM $MgCl_2$, 100 μM pppApG, 0.05 μCi/μl α-32P ATP. For molecular marker, both 5′ vRNA 1-14 and 70-mer mvRNA were 5′ dephosphorylated using the Antarctic Phosphatase (NEB) and radioactively labeled with T4 PNK (NEB) using γ-^{32}P ATP. All synthetic RNAs were purchased from Integrated DNA Technologies.

In vitro reactions were stopped by adding 2X RNA loading dye, heating 5 min at 95°C and immediately loaded on a 20% TBE 7 M urea-polyacrylamide gel. Each gel was exposed on a storage phosphor screen and read with an Amersham Typhoon scanner (Cytiva).

Electron microscopy: De novo RdRp elongation states were trapped by mixing 1.2 μM of H7N9 RdRp with (i) 1.2 μM 70-mer mvRNA with template:product stabilization; (ii) 6 μM hANP32A; (iii) 16 μM pS5 CTD in a final buffer containing 50 mM HEPES pH 8, 150 mM NaCl, 2 mM TCEP, 1 mM $MgCl_2$, ATP/GTP/CTP at 100 μM each, and 100 μM of non-hydrolysable UpNHpp (Jena Bioscience). The mixture was incubated for 4 hours at 30°C, centrifuged for 5 min at 11,000*g* and kept at 4°C before proceeding to grids freezing.

For grid preparation, 1.5 μl of sample was applied on each side of plasma-cleaned (Fischione 1070 Plasma Cleaner: 1 min 10 s, 90% oxygen, 10% argon) grids (UltrAufoil R 1.2/1.3, 300 mesh, Au). Excess solution was blotted for 3 sec, blot force 0, 100% humidity, at 10°C, using a Vitrobot Mark IV (ThermoFisher) before plunge freezing in liquid ethane.

Cryo-EM data collection: Automated data collection was performed on a TEM Titan Krios G4 (ThermoFisher) operated at 300 kV, equipped with a cold field emission gun, fringe-free illumination, a Falcon 4i direct electron detector camera, a Selectris X energy filter, and EPU software (ThermoFisher). Coma and astigmatism correction were performed on a carbon grid.

TIFF movies were recorded in counting mode at ×165,000 magnification, giving a pixel size of 0.73 Å, with a defocus range from −0.8 to −2.0 μm in 0.2 μm steps. Movies were fractioned in a total of 36 fractions (1.15 $e^-/Å^2$/fraction) for a total exposure of ~41 $e^-/Å^2$.

Image processing: A total of 29,627 movies were collected. Movie drift correction was performed using Relion's MotionCor implementation with all movie frames (*91*). All additional initial image processing steps were performed in cryoSPARC v4.5 (*92*). Contrast Transfer Function parameters were determined using the "Patch CTF Estimation" job. Realigned micrographs were manually inspected, and low-quality images were discarded, resulting in 28,109 micrographs retained.

Particles were automatically picked using a circular blob with a diameter ranging from 110 to 140 Å and extracted using a box size of 420 pixels by 420 pixels, Fourier-cropped to 200 pixels by 200 pixels. Successive two-dimensional (2D) classifications were used to eliminate particles displaying poor structural features. From a subset of one million particles, five ab initio reconstructions were generated. The best initial reconstruction was used as a common initial model to perform multiple 3D classifications using "heterogeneous refinement" jobs

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to remove particles displaying poor structural features. Remaining particles were grouped, Fourier-cropped to 300 pixels by 300 pixels, and subjected to a second round of 3D classifications to isolate particles in elongation states. The resulting 4,028,139 particles were subjected to "non-uniform refinement" jobs followed by multiple 3D classifications without alignments to further polish the particle selection.

A total of 1,580,146 particles displaying a clear template-product duplex within RdRp cavity were retained, Fourier-uncropped, and subjected to a "non-uniform refinement" job. On the basis of the estimated particle angles and shifts, a final "3D classification" job was performed. For each relevant RdRp conformation, particles were grouped and subjected to a final "non-uniform refinement." Post-processing was performed in cryoSPARC using an automatically determined B-factor. For each final map, the reported global resolution is based on the $FSC_{0.143}$ cut-off criterion. Local resolution variations were estimated in cryoSPARC.

Model building and refinement: A previous structure of elongating H7N9 RdRp was used as a starting point for model building. As bases were not resolved in the stem density, a model of the double stem region (nucleotides 16–29 and 51–62) was made using AlphaFold3 (*54*) that could be convincingly rigid-body fitted into the density. The atomic model was improved through iterative rounds of manual model building with COOT (*93*) and real-space refinement using Phenix with Ramachandran restraints (*94*). Validation was performed using Phenix. Model resolution according to the cryo-EM map was estimated at the 0.5 FSC cutoff. Structural analysis was performed in COOT and Chimera (*95*). Figures were generated using ChimeraX (*96*).

Conservation of the 1043–1048 cRNA structure in H5 HPAIV precursors

To assess the relevance of the 1043–1048 cRNA structure for LPAIV-to-HPAIV conversion in nature, we investigated whether it was present in LPAIV precursors of known H5 HPAIV outbreaks. We focused only on the 11 H5 LPAIV-to-HPAIV conversions resulting from MBCS acquisition through insertion and for which the nucleotide sequence was available (*1*). Viruses from these outbreaks can be sorted into three lineages, the geographical AEO and American lineages and a sublineage of the American lineage first isolated in Mexico in 1994 (Mx/94). Viruses from the Mx/94 lineage have evolved separately and stand out from other American H5 LPAIVs by their continuous circulation in poultry. LPAIV precursors have been identified only in 3 out of the 11 HPAIV outbreaks: Canada 1966, Mexico 1994, and South Africa 2006 (*1*). For all other outbreaks, a putative precursor was identified by using the HPAIV HA as nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) query against the nr/nt database and selecting the genetically closest HA sequence bearing a conserved RETR LPAIV cleavage site. For each LPAIV-to-HPAIV conversion event, one HPAIV representative was chosen. For each putative precursor, the predicted cRNA RdRp-trapping structure at the position equivalent to 1045 in NL_{RETR} was determined using our transient structure prediction tool described above.

Detection of putative NHR events

All consensuses containing complex insertions of nine or more nucleotides containing at least one C or U nucleotide (in cRNA sense) were identified. Sequences were aligned to the corresponding reference and selected as putative NHR events if the insert could not be explained by repeated duplication insertions. Putative NHR inserts were analyzed by nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) against the nr/nt database restricted to influenza A virus (NCBI:txid11320) or human (NCBI:txid9606) sequences to determine the putative origin of the insertion.

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ACKNOWLEDGMENTS

We express our continued respect and admiration for our beloved colleague Mathis Funk (30 October 1990–28 November 2024), a young and talented researcher who left our world too soon. This work was his passion; it is now his legacy. This article is dedicated to his memory. We thank M. Aguilar-Bretones, L. Bauer, B. Haagmans, S. Herfst, K. Schmitz, and S. Thewessen for their critical reading and suggestions. We thank S. Tureli for bioinformatics help and support. We thank A. Mehle for advice on the chicken polymerase I promoter cloning. The cryo-EM data were acquired at the CM02 CRG beamline operated by Institut de Biologie Structurale (IBS) at the European Synchrotron Radiation Facility in Grenoble. The purchase of this microscope was funded by the EquipEx+ France Cryo-EM project (ANR-21-ESRE-0046). We thank G. Schoehn for the establishment of the IBS cryo-EM facility. M. Peiris, Univ. of Hong Kong, provided A/Indonesia/5/2005 with permission from I. Kandun of the Indonesian government. A/mallard/Netherlands/3/1999 was obtained by the Erasmus University Medical Center, and A/mallard/Sweden/81/2002 was obtained by the Erasmus University Medical Center in collaboration with Umeå University (Sweden). **Funding:** This work was supported by ZonMw Off Road grant 04510012010056 (M.F.); Nederlandse Organisatie voor Wetenschappelijk Onderzoek ENW XS grant OCENW.XS22.1.121 (M.F.); National Institutes of Health/National Institute of Allergy and Infectious Diseases grant DP2 AI75474 (A.J.W.t.V.); Nederlandse Organisatie voor Wetenschappelijk Onderzoek ENW M1 grant OCENW.M.21.150 (M.R.); Horizon 2020 research and innovation program DELTA-FLU grant 727922 (M.R.); National Institutes of Health/National Institute of Allergy and Infectious Diseases contract numbers HHSN272201400008C and 75N93021C00014 (R.A.M.F., M.R.); National Institutes of Health/National Institute of Allergy and Infectious Diseases R01 award 1R01AI177487 (M.R.); and EMBL core funding. **Author contributions:** Conceptualization: M.F., M.R.; Methodology: M.F., M.I.S., B.A., A.C.M.d.B., S.C., A.J.W.t.V., M.R.; Investigation: M.F., M.I.S., R.M.H., B.A., P.J., T.M.B.; Formal analysis: M.F., B.A., M.R.; Software: M.F., A.P.G., A.J.W.t.V.; Visualization: M.F., B.A.; Funding acquisition: M.F., R.A.M.F., A.J.W.t.V., M.R.; Project administration: M.R.; Supervision: M.R.; Writing – original draft: M.F., M.R., B.A., S.C.; Writing – review & editing: M.F., M.I.S., R.M.H., B.A., P.J., T.M.B., A.C.M.d.B., A.P.G., R.A.M.F., S.C., A.J.W.t.V., M.R. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All data generated in this study are free to access. Next-generation sequencing data that underlie the findings of this study have been deposited in Dryad (*97*). The cryo-EM data from CM02 are available in the European Synchrotron Radiation Facility data portal (*98*). The coordinates and EM maps generated in this study have been deposited in the Protein Data Bank and the Electron Microscopy Data Bank (see table S3) under the following accession codes: PDB ID 9RAF and EMDB-53876 (*99*) and PDB ID 9RAG and EMDB-53877 (*100*). Code used to predict RdRp-trapping RNA structures and to analyze CirSeq results is available as an archived version in Zenodo (*69*) or in GitHub (*70*). Materials are available upon request, subject to a material transfer agreement, from the corresponding author: m.richard@erasmusmc.nl. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

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Figs. S1 to S9; Tables S1 to S3; MDAR Reproducibility Checklist; Data S1 to S7

Submitted 11 July 2024; resubmitted 17 July 2025; accepted 21 November 2025

10.1126/science.adr6632

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Polymerase trapping as the mechanism of H5 highly pathogenic avian influenza virus genesis

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Science **391** (6790), eadr6632. DOI: 10.1126/science.adr6632

Editor's summary

High-pathogenicity influenza viruses have not only been devastating the planet's wild and domestic bird populations, but they also represent a persistent threat of initiating a fatal human influenza pandemic. Funk *et al*. investigated how recombination of low-pathogenicity viruses is prone to incorporating sequences for the furin multibasic cleavage site in the hemagglutination gene, which promotes cell invasion by virus. The authors found that transient RNA structures in the virus replication machinery caused it to stutter on adenine-/uridine-rich sequences and allowed the nucleotide insertions that translate into the cleavage site. The authors suggest that these sorts of transient structures might also be present in other RNA viruses. —Caroline Ash

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